
GOVERNMENT OF ANDHRA PRADESH

Chief Minister's Office - AP IAS Performance Tracking System

COMBINED SCORE - 6 of 6

Professional Performance Index

The single overall score - a normalised, weighted blend of the five component scores - with a fair, state-wide ranking among officers who share the same set of scores.

e-Office + Swarnandhra KPI + Gol Funds + Public Perception + GSDP

A plain-language note - every formula matches the live calculation.

1 What this score measures

Each of the five component scores looks at one slice of an officer's work. To compare officers overall, those slices must be brought together into one number out of 100 - the Professional Performance Index (PPI). The five components are e-Office, Swarnandhra KPI, GoI Funds, Public Perception and GSDP. Each is produced by its own method (see its own note); the PPI is purely about COMBINING them fairly - which raises three real problems it must solve:

- the five may sit on slightly different spreads, so they must be put on one ruler first;
- not every officer has all five (a new posting may have no survey yet), so the blend must cope with missing pieces without punishing the officer;
- comparing an officer with five scores against one with two would be unfair, so ranking must be done within like-for-like groups.

Putting different measures on one 0-100 ruler (normalisation)

Different scores are measured in different units - rupees crore, minutes per file, survey points, percentages. You cannot fairly average rupees with minutes. So each measure is stretched onto a common 0-to-100 ruler using MIN-MAX normalisation:

MIN-MAX NORMALISATION

$$\text{normalised} = (\text{value} - \text{lowest}) / (\text{highest} - \text{lowest}) \times 100$$

lowest = the smallest value across all officers

highest = the largest value across all officers

After the stretch, the best officer on that measure scores exactly 100, the weakest scores 0, and everyone else keeps their exact place in between. The only assumption is a straight-line stretch between the two ends, which is transparent and easy to audit. If every officer happens to be identical, the rule would divide by zero, so in that case all are simply set to 100.

The weighted average - letting important pieces count more

Most scores blend several pieces of evidence. A weighted average lets the more important pieces count more. Give each piece a weight, multiply each piece by its weight, add them all up, and divide by the sum of the weights used.

WEIGHTED AVERAGE

$$\text{weighted average} = \frac{w1.v1 + w2.v2 + \dots + wn.vn}{w1 + w2 + \dots + wn}$$

v = a 0-100 piece (sub-score), w = its weight (importance).

A weight of 0.45 means that piece decides 45% of the result; a weight of 0.10 means only 10%. Dividing by the SUM of the weights actually used is important - it is what lets a missing piece be skipped without distorting the rest.

2 Step 1 - put the five on the same ruler

Although every component is already 0-100, the spread of values can differ. So each component is min-max normalised across all officers, making the best officer on each component exactly 100 and the weakest 0. This guarantees that, under equal weights, each component truly contributes equally - no component dominates merely because its raw numbers happen to be larger or more spread out.

A Normalised / Absolute switch lets a viewer instead see the raw 'Absolute' component scores; in that mode the components are blended as-is, without re-stretching. The default view is Normalised.

3 Step 2 - the weighted average over the AVAILABLE scores

The PPI is the weighted average of ONLY the components the officer actually has. By default every component carries equal weight (so the PPI is a simple average of whatever exists), but the administrator can set different weights in the Score-Weights page - for example to make e-Office count double.

PROFESSIONAL PERFORMANCE INDEX

$$\text{PPI} = \frac{\text{SUM (weight_i x score_i)}}{\text{SUM (weight_i)}}$$

summed ONLY over the
components the officer
actually HAS

components i = e-Office, Swarna, GoI, Perception, GSDP
default weight_i = 1 for every component (a simple average)

In plain words: an officer is never penalised for a score that does not exist yet - the index is the fair average of whatever they DO have. Dividing by the sum of the weights USED is what makes a missing component harmless.

4 Step 3 - ranking 'within the combination'

Ranking an officer who has all five scores directly against one who has only two would be unfair - more scores simply means more chances to be high or low. So officers are grouped by WHICH set of scores they possess - their 'combination' - and the rank shown is the officer's position, state-wide, among everyone who has EXACTLY the same combination.

The dashboard lets you pick a combination from a long list - for example 'All 5 Scores', 'Except GoI Funds', or a pair like 'e-Office + GSDP' - and it ranks officers only within that group. A coverage matrix on the report page shows how many officers fall into each combination.

5 Worked example

WORKED EXAMPLE - Professional Performance Index

Officer's normalised component scores:

e-Office 80, Swarna 74, GoI 85, Perception 70, GSDP 83

Equal weights (all = 1):

$$\text{PPI} = (80 + 74 + 85 + 70 + 83) / 5 = 392 / 5 = 78.4$$

Administrator raises e-Office weight to 2 (others stay 1):

$$\begin{aligned} \text{PPI} &= (2 \times 80 + 74 + 85 + 70 + 83) / (2 + 1 + 1 + 1 + 1) \\ &= (160 + 312) / 6 = 472 / 6 = 78.7 \end{aligned}$$

If this officer had only e-Office + GSDP (a 2-score combination):

$$\text{PPI} = (80 + 83) / 2 = 81.5$$

...and they are ranked only against others who also have exactly those two.

6 The statistics, explained

Why normalise before averaging

Averaging numbers on different spreads silently lets the widest-spread component dominate. Normalising every component to an identical 0-100 spread removes that bias, so equal weights really do mean equal influence.

A weighted average over a variable set of inputs

The key subtlety is dividing by the SUM OF THE WEIGHTS ACTUALLY USED, not by a fixed 5. That is what lets an officer with three scores be judged on the fair average of those three, neither rewarded nor punished for the two they lack.

Like-for-like ranking

Grouping by the combination of available scores keeps the ranking honest when data coverage differs between officers. It is the same principle as comparing students only within the same set of subjects they sat.

Normalised versus Absolute

Normalised view answers 'where does this officer stand relative to peers on each component?'; Absolute view answers 'what did this officer actually score?'. Both are offered; the index and rank shown follow whichever view is selected.

7 Reading it on the dashboard & FAQ

The Professional Performance Index page shows, for each officer, the five component scores (each with its own rank in the group), the final Index, and the state-wide Rank within the selected combination. Clicking a component jumps to that score's full page with the officer highlighted. The report PDF includes a coverage matrix of all combinations.

Q. I only have three of the five scores. Am I being penalised?

A. No. The index averages only the scores you have (dividing by the weights used), and you are ranked only against others with the same three.

Q. Why did my Index change when I switched Normalised / Absolute?

A. Normalised stretches each component to 0-100 across peers; Absolute uses the raw component scores. The blend, and therefore the Index, differs between the two views.

Q. Can the weights be changed?

A. Yes - the administrator sets them in the Score-Weights page (default is equal weight for all five). Changing a weight re-blends the Index for everyone.

Q. Why rank 'within the combination' instead of one big list?

A. Because more available scores means more chances to be high or low; comparing only like-for-like groups keeps the ranking fair.

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Glossary of terms

Component score

one of the five inputs (e-Office, Swarna, Gol, Perception, GSDP).

Normalise (min-max)

stretch each component to 0-100 across all officers.

Weight

the importance given to a component (default 1 = equal).

Available scores

the components an officer actually has data for.

Combination

the exact set of scores an officer has; ranking is within the same set.

Absolute vs Normalised

a page switch to view raw component scores or the stretched ones.

PPI

Professional Performance Index - the combined, weighted, 0-100 score.

All weights, benchmarks and difficulty-factor floors/ceilings are stored in the score_config table and are configurable by the administrator in the Score-Weights page of the dashboard, so the system can be re-tuned without changing any code. Scores recompute automatically whenever the underlying data changes. Every formula in this note matches the live calculation exactly.